

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Phase #: 2				
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Pauli Correlation Encoding is All You Need!

A Hybrid Architecture for Power-Grid Storage Planning via a Pauli-Correlation-Encoded Conditional Generative Quantum Eigensolver

Abstract— We present a theoretical framework of our solution with a plausible experimentation code base where we have designed a full hybrid workflow to solve the optimization task of storage *siting and sizing* for resilience and AI load integration, modeled as a *stochastic mixed-integer nonlinear* problem. The pipeline consists of a **Quantum Master Solver** that addresses the combinatorial search problem through a novel redesigned **Conditional Generative Quantum Eigensolver** (cGQE), to which we have introduced **Pauli Correlation Encoding** (PCE) that drastically reduces the number of necessary qubits. A classical solver enforces the nonconvex *Alternating Current* (AC) physics through a training signal, iterating until convergence. This architecture offers a **context-aware, generalizable** solution that can be trained once, after which the model generates a suitable circuit for a new problem instance in a single forward pass without retraining. This saves both significant **re-training costs** and the **number of qubits** required, making the approach suitable for NISQ-era hardware.

1 Introduction

This work is part of Phase II of the Global Industry Challenge under the United States Department of Energy track. AI data centers and new industrial loads are stressing power grids everywhere which may directly be a vulnerability that compromise national security. This is creating an urgent planning question: *where* to place battery energy storage systems (BESS) and microgrids, and *how large* to make them, so that the grid remains resilient under uncertainty. This is a hard combinatorial problem where very candidate bus can either host a system or not, and size choices multiply the options. On top of that, power flows in a nonlinear way, and planners must consider a scenarios tree of weather realizations and $N-1$ contingencies.

Classical optimization tools work well for small networks, but they slow down dramatically as the network and scenario list grow. Quantum computing offers a way around this wall, yet most quantum methods need one qubit for every binary decision which makes them hardware-intractable on current NISQ-era devices when the problem scales

We solve both issues at once. Our design is a hybrid split. A quantum generative model handles the combinatorial siting and sizing decisions. A classical solver handles the nonlinear power flow physics. Inside the quantum model, we embed Pauli Correlation Encoding [1] into the Conditional Generative Quantum Eigensolver [2]. This compression lets us compress m binary variables onto $n \approx m^{1/k}$ qubits while inheriting plateau-resistance by construction. The result is a single trained model that can instantly suggest a good circuit for a brand new planning scenario without retraining, saving both time and hardware.

2 Problem and Decomposition

We split the problem into two parts. The first part is the discrete investment master. It only picks sites and sizes. It has no continuous physics, but it is combinatorially hard. The second part is the continuous operations subproblem. For a fixed plan, it finds feasible power flows and checks every scenario. We link the two parts in a loop where the quantum master proposes plans and the classical subproblem evaluates them and sends feedback, which reshapes the quantum-master's goal for the next round.

3 Architecture

Figure 1 shows the full pipeline and we detail each module below.

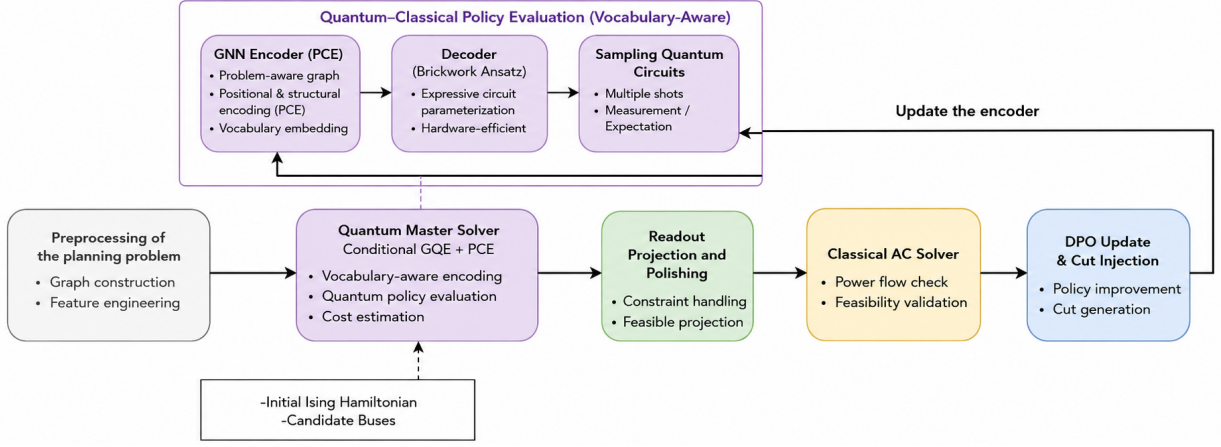


Figure 1: Full pipeline of the proposed hybrid quantum-classical system. Module colors correspond to: **gray** = classical preprocessing, **purple** = quantum master SOLVER (Conditional-GQE-PCE), **green** = readout & projection, **yellow** = classical AC subproblem, **blue** = DPO update & cut injection.

3.1 Module 1: Scenario generation and convex preprocessing

Experiments are conducted on the IEEE 14-, 30-, and 57-bus systems under projected load growth, renewable variability, and $N-1$ contingencies. A preliminary DC-OPF (or SOCP relaxation) ranks buses by sensitivity, pruning the candidate set before quantum encoding and reducing the number of decision variables m (and qubits n). Size buckets are fixed, and the initial Ising Hamiltonian H_0 incorporates investment costs, budget constraints, and mutex penalties. This preprocessing stage is the most effective mechanism for controlling qubit requirements.

3.2 Module 2 — Quantum master (cGQE with PCE)

Each candidate site-size pair is mapped to a binary variable. Compact log encoding is used for size classes, while budget and mutex constraints are added as penalty terms, producing a QUBO over m variables.

The key compression mechanism is **Pauli Correlation Encoding (PCE)**. Instead of assigning one qubit to each variable, every variable is represented by a k -body Pauli correlation,

$$\mathcal{O}_i = \bigotimes_{j=1}^n P_{ij},$$

allowing $m = O(n^k)$ variables to be encoded on only n qubits. Binary decisions are recovered from the sign of the measured correlations. To reduce hardware overhead, observables are grouped into three commuting families, requiring only **three measurement settings** regardless of m .

A conditional generative decoder forms the core of the quantum master. A transformer, conditioned on a GNN representation of the grid, generates circuits token-by-token. The decoder vocabulary is restricted to brickwork-PCE tokens, ensuring every generated circuit is a valid PCE ansatz. A deterministic assignment module computes the Pauli strings and correlation mapping and passes this information to the decoder, enabling it to learn which variables correspond to which correlations.

Training is performed with **Direct Preference Optimization (DPO)**. Candidate circuits are ranked using their true total cost from the classical evaluation stage, and the decoder is updated to increase the likelihood of better circuits. Since DPO relies only on circuit rankings and model probabilities, no gradient passes through the quantum hardware. Once trained, the model is amortized: new scenarios require only a single forward pass rather than a new variational optimization loop.

3.3 Module 3 — Readout and confidence-weighted projection

After measurement, the sign of each correlation gives the binary decision, while its magnitude provides a confidence score. Since the raw bitstring may violate budget or mutex constraints, we solve a small projection ILP that finds the nearest feasible solution while preserving high-confidence decisions and adjusting low-confidence ones. A brief local search further improves the result. This lightweight step guarantees that every candidate passed to the classical evaluation stage is feasible.

3.4 Module 4 — Classical subproblem and cut generation

For each feasible investment plan, we solve a security-constrained optimal power flow across all scenarios. A DC-SCOPF is used in Phase II for efficiency, while an SOCP relaxation provides a richer AC representation. Infeasible cases generate feasibility cuts and feasible cases generate optimality cuts, which are translated into additional terms in the quantum master’s Hamiltonian. This stage handles network physics, $N-1$ security, and scenario uncertainty using classical optimization tools.

3.5 Module 5 — DPO update and cut injection

Candidate solutions are ranked by their true total cost, and the decoder is updated using DPO. Before the next iteration, Hamiltonian coefficients are rescaled and inactive cuts are removed to maintain numerical stability. The updated cuts are injected into the quantum master, and the process repeats until the cost–resilience tradeoff converges.

4 Stakeholder Relevance

The main beneficiaries are **transmission planners**, **independent system operators (ISOs)**, and **distribution utilities** planning storage deployment under uncertain future demand. The method addresses a practical question: given a budget and a 10–20 year planning horizon, which buses should receive storage and at what capacity to maximize resilience while controlling cost.

5 Platform Justification and Resource Needs

Our workflow is hybrid, so we need a platform that lets classical and quantum code run side by side. We choose the **qBraid Lab** environment for integration because it offers combined CPU, GPU, and QPU access in one place. This lets our PyTorch decoder, classical power flow solver, and quantum circuit execution share data smoothly.

For simulation and algorithm development we use the **qBraid QIR simulator** and **NVIDIA GPU accelerated simulation** via Qiskit GPU and cuQuantum. These give fast statevector checks for up to tens of qubits and circuit depths in the low hundreds, which is exactly the range we need for debugging the PCE ansatz.

For physical hardware validation we target **IBM Heron r1**. Our compressed circuits need only 5 to 7 qubits and depths of 2 to 4 layers, well within Heron’s specifications. We request 3×1024 shots per circuit, one batch for each commuting Pauli family. If error rates require it, we can layer **Fire Opal error mitigation software** on top of the IBM runs.

6 Validation and Current Status

We have checked every module for consistency. The classical loop alone reproduces the MISOCF baseline on the IEEE 14 bus system, confirming the subproblem and cut logic. The PCE encoding, three basis readout, and confidence projection have been unit tested on small Ising examples. A known optimal bitstring was encoded, measured, and recovered correctly through the pipeline. The DPO training loop was verified on synthetic problems of similar size, showing a capability learning without gradients through the quantum step.

The complete prove of concept codebase is made available at <https://github.com/AshrafBoussahi/GIC-2-DOE> at the time of submission.

7 Conclusion

We have presented our approach, a hybrid quantum-classical architecture for the strategic siting and sizing of battery storage under multi-scenario uncertainty and AC power flow constraints. The central technical contribution is the integration of Pauli Correlation Encoding into the Conditional-GQE framework via a restricted brickwork vocabulary and DPO preference training, yielding a generative model that produces compressed, plateau-resistant circuits for unseen planning instances in a single forward pass. The decomposition is principled: combinatorial search is handled by the quantum master, nonconvex physics by a mature classical SCOPF solver. The architecture is NISQ-ready today ($n = 5-7$ qubits, 3 measurement settings), classically benchmarkable on equal terms, and has a defensible scaling path to Phase III contingent on measured performance.

References

- [1] M. Sciorilli et al., “Towards large-scale quantum optimization using a hybrid approach based on Pauli Correlation Encoding,” *arXiv:2501.06241*, 2025.
- [2] S. Minami et al., “Generative quantum combinatorial optimization by means of a novel conditional generative quantum eigensolver,” *Digital Discovery*, vol. 4, no. 8, pp. 2229-2243, 2025.
- [3] B. Nakano et al., “Generative Quantum Eigensolver (GQE) and its application to ground state search,” *arXiv:2401.09253*, 2024.
- [4] K. Panaganti et al., “Conditional Generative Quantum Eigensolver,” *arXiv:2411.03555*, 2024.
- [5] R. Rafailov et al., “Direct Preference Optimization: your language model is secretly a reward model,” *NeurIPS 2023*.
- [6] C. Barrows et al., “The IEEE Reliability Test System: a proposed 2019 update,” *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 119-127, 2020.
- [7] M. Sengupta et al., “The National Solar Radiation Data Base (NSRDB),” *Renew. Sustain. Energy Rev.*, vol. 89, pp. 51-60, 2018.